Compilation Regime Geometry: A Receipted Programme for Mapping Exact Optimization Families

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Abstract

Compiler optimization is usually evaluated by the best cost found on a benchmark set. That endpoint does not reveal where a simple donor family is already exact, which structural trades buy improvement, how large an exact search family must be, or whether regime membership can be predicted from input structure. We call this collection a *compilation regime geometry* and operationalize a receipted mapping programme with five components: donor-optimal region, exact trade/refutation set, sufficient normal-form/search bound, structural membership or cost rule when one exists, and prospective validation. The programme is deliberately falsifiable component by component. In shared-Tag TARE, the unit objective has an all-n support-two normal form, while successive closed forms are refuted by split, borrow, phantom-borrow, and hybrid configurations; an objective-indexed theorem later identifies a cone in which support two remains sufficient. In the rank-2 companion grammar, an initial support-five theorem is progressively tightened to the exact intrinsic support number one. In SixLCU, a pair-derived donor-exact predicate is upgraded to an all-instance theorem even though optimal witnesses can be globally large. A third-family StabPrep transfer preserves the trade-mapping programme and sharpens the cross-family motif question. In the originally frozen natural feature vocabulary, mixed cells impose an irreducible 43/1,146 floor. Under a prospectively frozen enlarged vocabulary, all 1,146 states with $n \leq 3$ are exactly separated, and exhaustive search proves that four features are necessary and sufficient. That compact law fails on unseen $n = 4$ states and requires no hypothesized state-sign feature, so transfer and mechanism attribution remain unsupported. Thus the transferable object is the mapping discipline together with a domain-local boundary law, not a universal one. The results are machine-checked or finite-domain exact under their stated grammars/objectives and carry no physical quantum-advantage claim.

Keywords: quantum compilation; exact optimization; normal forms; phase diagrams; resource forecasting; machine-checked analysis.

1 Introduction

A compiler benchmark answers a practical question: which implementation has lower cost on the tested inputs? A regime map asks a structural one: on which inputs is the simple incumbent already exact, what changes when it is not, what restricted family is sufficient for an exact optimum, and can those regions be recognized without rerunning the full optimizer? These questions turn optimization output into an object of study.

We use *compilation regime geometry* for that object and make it operational rather than metaphorical. A map has up to five separately evidenced components: a donor-optimal region;

exact counterexamples/trade configurations that leave it; a sufficiency or normal-form theorem reducing unrestricted search; a structural classifier/cost expression if the family admits one; and prospective tests on frozen new inputs. A component may be refuted without invalidating the others.

That last clause became essential as the programme matured. The founding TARE case initially suggested a compact two-trade map. Later exact tests found a third phantom-borrow configuration and then a fourth split+borrow hybrid, while the all-n support-two theorem remained untouched. The third-family StabPrep study delivered an even stronger warning: the mapping programme transfers, but a clean low-order boundary does not.

The present synthesis therefore asks which parts of regime geometry are genuinely transferable. We report current results across TARE/R6M, a rank-2 shared-Tag grammar, SixLCU, and StabPrep, together with objective-indexed and production-semantics stress tests. The contribution is a receipted analysis programme plus the cross-family facts it has actually earned, not a claim that all compiler families share one geometric form.

2 Methods: constructing a regime map

2.1 Donor region

Each family starts with a frozen incumbent/donor definition and an exact referee where feasible. The donor-optimal region consists only of inputs for which exact equality is established. A donor tie is not credited as a new method.

2.2 Trade discovery by refutation

Before testing a proposed closure, the restricted family and hostile panel/generator are frozen. A strict exact-referee gap serializes a witness and becomes a trade/refutation class. Later repairs must preserve the earlier counterexample and are tested against new hostile constructions.

2.3 Sufficiency and intrinsic support

A finite search bound is promoted only after local machine checks are composed into an all-n statement. A first valid bound is not assumed tight. Subsequent ladders may lower it; the *intrinsic support number* is the smallest exact bound actually proved.

2.4 Structural classification and forecasting

A regime predicate or static cost expression is frozen before its evaluation panel. Zero finite-panel error establishes only that panel result unless an all-instance proof is supplied. A classifier can be refuted while a broader exact normal form remains valid.

2.5 Objective indexing

A map is indexed by both grammar and objective. Reweighting central/noncentral frames, Restore cost, Tag cost, or rotation terms can change the set of profitable trades and the required support. Objective transfer therefore requires a separate theorem or counterexample search.

2.6 Prospective and hostile validation

Whenever a structural rule is proposed, the programme prefers a prediction digest or frozen hostile generator before exact-referee access. Honest terminals include confirmation, refutation, donor absorption, partial proof, and cannot-check. Every result is bound to a protocol/receipt and later replayed or independently reconstructed.

3 Results

3.1 TARE: exact support, evolving trade vocabulary

For the frozen R6M unit support-count objective, R6S proves an all-n support-two normal form. That theorem survives every later regime-classifier failure. QG-5 refutes the original three-family cost formula on a fresh $n = 3$ instance; QG-5b enlarges the borrow family and restores zero error on 9,547 compared instances; QG-7 then finds 64 exact hybrid witnesses outside that repair; QG-7b covers the new configuration and closes 10,481 verified rows; QG-7c reduces the all-n closed-form classification to one pinned comm-$s2$ lemma-open sector. The map therefore has an exact normal-form layer and a still-open explanation layer.

QG-8 adds objective structure. Support at most two is sufficient for all n throughout the half-space cone

$$t_c \geq 2t_r \quad \text{and} \quad t_{nc} \geq 2t_r.$$

The baseline unit objective lies on the central hyperplane. The coefficient-reweighted O1 objective sits outside and contains the previously observed support-three witness, so support-two exactness is not an objective-free property.

3.2 Rank-2 grammar: from a valid bound to an intrinsic one

Wave 1 established a finite all-n support bound of five for the R6I rank-2 grammar. The QG-9 proof ladder subsequently tightens the bound through four, three, and two to support at most one. Its V6 result proves $C_{DP} = C_{\text{cap1}}$ for every n, so the current intrinsic support number is $\kappa_{R6I} = 1$. The key proof step relocates the shared Tag after localizing each block to one anticommuting core, a transformation missing from the earlier per-block syndrome-preserving proof grammars.

QG-6 provides a useful contrast. Production transition semantics infer a rewrite-relevant syndrome rank of five for R6I, which is a sound route to the earlier finite bound but not tight compared with $\kappa = 1$. For R6M the inferred rank is two, matching its earned support-two theorem. Production rank is therefore an informative theorem-discovery signal, not a universal formula for intrinsic support.

3.3 SixLCU: a low-order boundary theorem

The SixLCU transfer first identifies exact donor failures and a trade ladder using the family's finite referee. QG-12 upgrades the empirical pair-derived predicate to a theorem: for every admitted batch and every n, the donor cost C_F equals the unrestricted cost C_U if and only if P0 holds. The boundary certificate is low-order even though optimal witnesses can require larger/global block structure. This separates certificate arity from witness arity.

3.4 StabPrep: transfer with a first-class structural refutation

QG-15 applies the same five-component workflow to stabilizer-state preparation with H/S/SDG cost 1 and CNOT cost 3, using complete exact state graphs of 6, 60, 1,080, and 36,720 states for $n = 1$ through 4. The donor-exact region contracts sharply with n; four trade classes (ORDER, PIVOT, ROUTE, GLOBAL) are witnessed, and a prospective held-out $n = 4$ forecast is refuted rather than post-hoc repaired.

The originally frozen natural positive-conjunction vocabulary does not determine donor exactness. Its 12 mixed feature cells impose an irreducible error floor of 43/1,146, irrespective of predicate budget. A prospectively frozen enlarged vocabulary changes the within-domain conclusion: across all 1,146 states with $n \leq 3$, 127 features form 1,109 cells with no mixed cell. Exhaustive rejection of every one-, two-, and three-feature subset, together with a verified four-feature witness, proves that the minimum exact separator has size four.

The favourable separator is nevertheless local and mechanism-agnostic. Its witness uses two original-vocabulary and two donor-path coordinates but no state-sign coordinate, while any two feature blocks already attain zero floor. The proposed state-sign mechanism is therefore not supported. The separator also fails on the unseen $n = 4$ panel, where it makes 32/120 errors, matching the parent lookup baseline and a shuffle null ($p = 0.51$). StabPrep thus supports a domain-local low-order boundary law, not cross-size transfer or a universal low-order boundary claim.

4 Discussion

The cross-family result is asymmetric. The mapping *questions* transfer well: incumbent region, exact trade witnesses, search/normal-form complexity, structural recognizability, and prospective prediction can all be posed for materially different compilers. The *answers* need not share a form. TARE has a small exact support normal form but an evolving closed-form trade vocabulary. R6I collapses further to support one. SixLCU admits a strikingly low-order exact boundary. StabPrep does not admit an exact boundary in its frozen natural feature vocabulary at all.

This suggests that regime geometry should be treated like a phase-diagram methodology rather than a theorem schema. A useful map can contain a negative component: 'no clean predicate in this vocabulary' is itself structural information, especially when mixed feature cells prove that more search over the same vocabulary cannot fix the classifier.

The support-bound results similarly distinguish validity from tightness. QG-6's production-inferred rank can guide theorem discovery, but QG-9 demonstrates that a semantics-derived finite bound may be loose by a large factor. Tightening a bound requires new transformations, not more confidence in the first certificate.

Objective dependence provides the final guardrail. A geometry belongs to a grammar–objective pair. QG-8 turns that guardrail into a theorem for one objective cone; outside the cone, a support-three control already exists. Comparisons across objectives should therefore report map transitions rather than assuming one canonical geometry.

5 Related work and boundary

The mapped compiler families sit inside well-established quantum-compilation traditions. The TARE case descends from a stabilizer-formalism Pauli block-encoding construction [1], while anticommuting-unitary partitioning predates the present regime analysis [2]. Pauli-level compiler optimization and global binary-symplectic simplification are also established directions [3, 4].

The StabPrep transfer relies on the standard stabilizer-circuit setting in which efficient tableau simulation and canonical forms are classical results [5]. Dynamic programming, Pauli/symplectic representations, stabilizer synthesis, and family-specific donor constructions therefore receive no novelty credit merely for appearing in this programme.

The bounded candidate residual is methodological and cross-family: making donor regions, exact trade refutations, sufficiency/tightness, structural recognizability, and prospective prediction into separately receipted components that can succeed or fail independently. The current manuscript does not claim that no earlier work maps any compiler family in this way. A fresh literature closure must compare the exact composition against normal-form, exact synthesis, phase/regime, and certified resource-estimation literature before submission.

The StabPrep result is particularly important for novelty discipline because it prevents the paper from selling an artifact of the first two families as a universal principle. Any field-level statement must include the negative transfer.

Mapping a problem space into regions is an established methodology. The idea that a problem space should be partitioned into regions, rather than summarised by a best score on a benchmark, is not new. Instance Space Analysis, built on Rice's algorithm-selection framework, selects the instance features most correlated with algorithm performance, projects them into a low-dimensional space, and identifies regions in which each algorithm performs well or poorly; a new instance is then routed by the region it falls into [6]. Christiansen and Smith-Miles apply it to the quadratic assignment problem and use it for a second purpose that bears directly on the argument here: the projection exposes which parts of the instance space the standard benchmark library fails to cover, and they find that the established library has poor coverage of a large region. Their response is to construct new instance classes to fill it.

That is the same objection this programme raises against evaluating compilation by best cost on a fixed benchmark set, and it has been made more rigorously there than here. The difference lies in what the regions are regions *of*. An instance space partitions by *relative algorithm performance*: which method wins, and by how much, measured empirically. The geometry mapped here partitions by *exactness*: whether a donor family is provably optimal on a region, which structural trades are refuted rather than merely outperformed, and how large a search family must be to be sufficient. A region where one metaheuristic reliably beats another is a different object from a region where a simple family is already exact and nothing can beat it, and only the second supports a refutation. The methodologies are complementary — a regime map says where to stop searching, an instance space says what to run when searching is still required.

6 Limitations

Most results are machine-checked in deliberately finite or algebraically frozen grammars. They do not establish how often the same structures occur in production-scale chemistry or hardware-constrained compilation. Exact referees are available only for families/scales where the state space can be exhaustively or proof-carryingly searched.

The map components are not statistically independent: later lanes are motivated by earlier refutations, although each result-bearing protocol is frozen before its own outcome. This makes the programme a sequential theorem/discovery process rather than a preregistered fixed battery.

'Structural predicate' always means predicate in a declared feature vocabulary. QG-15b proves non-separability only inside its frozen StabPrep vocabulary; a prospectively frozen enlarged vocabulary does recover exact determination on $n \leq 3$ with four features, but not on unseen $n = 4$, so the recovered law is domain-local rather than universal. Likewise, TARE's current finite trade vocabulary is not yet proved all-n complete.

Finally, exact structural cost is not physical runtime or quantum advantage. Architecture, routing, fault tolerance, state preparation, measurement, and other resource coordinates can reorder practical choices.

7 Conclusion

Compilation regime geometry is best understood as a falsifiable mapping programme, not as a promise that every compiler family has a simple phase boundary. Across TARE, R6I, SixLCU, and StabPrep, the programme repeatedly turns exact optimization into structural questions about donor regions, profitable trades, normal-form complexity, recognizability, and prospective prediction.

The strongest general lesson is negative in a productive sense. Small exact support, small boundary certificates, and clean closed-form forecasts are family- and objective-dependent properties. TARE, R6I, and SixLCU earn different kinds of compression; StabPrep explicitly refuses one of them. A regime map is valuable precisely because it records both the compression that can be proved and the structural complexity that survives.

8 Reproducibility

The QG programme freezes each lane under `development/orion-qg-regime-geometry/` and stores analyzers/results under `research/extensions/orion-qg/`. Founding TARE results additionally bind to the R6 receipts under `research/extensions/orion-q/`. The running wave-2 record names merge commits, protocol hashes, result digests, independent generic-verifier decisions, and replay status for the headline lanes.

A reproduction should verify the final theorem/refutation state, not only wave-1 snapshots. In particular, QG-9 V6 supersedes the earlier R6I support-5 bound; QG-12 upgrades SixLCU's predicate; QG-15/QG-15b refute the low-order-boundary motif; and QG-7 series refines TARE's closed-form classification without contradicting R6S.

Paper-level verification order is listed in `REPRODUCE.md`. Protocol/result chronology and the preserved protected-subject boundary are hard integrity checks.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. The principal integrity risks are post-outcome family expansion, theorem promotion from finite panels, selective omission of refutations, and accidental transfer of a theorem across objectives/grammars. The receipt/protocol discipline makes these boundaries explicit.

Some lanes use large exhaustive domains, including hundreds of millions of local configurations. Computational scale is a verification cost, not evidence of novelty or physical relevance. Protected chemistry subjects remain sealed unless a separate pre-outcome protocol releases them, and no result in this paper grants physical quantum-advantage or deployment authority.

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